

Tutorial 8 Answers

Problem 9-1

Pitch diameter = 4.8 in
Circular pitch = 62.83 in
Addendum = 0.2 in
Dedendum = 0.25 in
Clearance = 0.05 in
Tool thickness = 0.3142 in

Problem 9-5

The diametric pitch = 10 (for the largest standard tool size)

(a) Number of teeth: 25 and 80

(b) Number of teeth: 15 and 48

Problem 9-7

Number of teeth : 16 (pinion) and 96 (gear)

Pitch diameters: 3.2 in and 19.2 in

Contact ratio: 1.673

Problem 9-25

790.00 rpm, same direction of Gear 6's velocity

Problem 9-35

22.09 rpm, same direction of Gear A's velocity

Problem 9-39

979.6 rpm and 2742.9 rpm